



EMERGENT™

Configuration as Code Series:

CIS Benchmarks



Overview

- Highlight the tools available from RHEL to meet compliance needs
- Make quick work of those tools via Ansible
- Implementing it at scale with Ansible Automation Platform

Ansible Automation Platform

- **Red Hat Ansible Automation Platform (AAP)** is a comprehensive platform for IT automation, built on the foundation of open-source Ansible automation engine
- It provides enterprise-grade features for managing and scaling Ansible automation across teams, environments and hybrid IT environments
- AAP is comprised of various services that work together to meet your automation needs, providing the ability to **store, make decisions for, and execute automation**
- All features and capabilities are accessible through a unified **user interface (UI)** and **RESTful application programming interface (API)**





CIS Benchmark

- A community-driven nonprofit, responsible for the CIS Controls® and CIS Benchmarks®, globally recognized best practices for securing IT systems and data. We lead a global community of IT professionals to continuously evolve these standards and provide products and services to proactively safeguard against emerging threats. Our CIS Hardened Images® provide secure, on-demand, scalable computing environments in the cloud.
- <https://www.cisecurity.org/about-us>



CIS Benchmark Levels

- The Level 1 profile is considered a base recommendation that can be implemented fairly promptly and is designed to not have an extensive performance impact. The intent of the Level 1 profile benchmark is to lower the attack surface of your organization while keeping machines usable and not hindering business functionality.
- The Level 2 profile is considered to be “defense in depth” and is intended for environments where security is paramount. The recommendations associated with the Level 2 profile can have an adverse effect on your organization if not implemented appropriately or without due care.
- The STIG profile replaces the previous Level 3. The STIG profile provides all recommendations that are STIG specific. Overlap of recommendations from other profiles, i.e. Level 1 and Level 2, are present in the STIG profile as applicable.



OpenSCAP

- OpenSCAP is a suite of open-source tools and libraries designed for automated security auditing, vulnerability assessment, and configuration compliance checking of systems. It leverages the Security Content Automation Protocol (SCAP), a set of standards developed by NIST to enable consistent and repeatable security processes. OpenSCAP helps organizations ensure their systems adhere to security policies, identify potential vulnerabilities, and maintain a secure computing environment.
- <https://www.open-scap.org/>
- Red Hat maintains most major reports and remediations



Methodology



DEMO



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Command references

List available reports maintained by Red Hat

```
oscap info /usr/share/xml/scap/ssg/content/ssg-rhel8-ds.xml
```

Run a CIS Level 1 Report

```
oscap xccdf eval --profile xccdf_org.ssgproject.content_profile_cis_server_l1 --results ~/OpenScap_results_file.xml /usr/share/xml/scap/ssg/content/ssg-rhel8-ds.xml
```

Run a CIS level 2 Report

```
oscap xccdf eval --profile xccdf_org.ssgproject.content_profile_cis --results ~/OpenScap_results_file.xml /usr/share/xml/scap/ssg/content/ssg-rhel8-ds.xml
```

Generate a html report for CIS Level 2

```
oscap xccdf generate report --output ~/OpenScap_results_file.html ~/OpenScap_results_file.xml
```



Websites Referenced

Red Hat Official Ansible CIS Compliance Ansible roles:

<https://github.com/RedHatOfficial/ansible-role-rhel8-cis>

<https://github.com/RedHatOfficial/ansible-role-rhel9-cis>

CIS:

<https://www.cisecurity.org/about-us>

Red Hat on demand Ansible Labs:

<https://www.redhat.com/en/interactive-labs/ansible>